

Careerspherex

1. Functionality :

- 1. User Registration & Login**
- 2. User Profile Management**
- 3. Skill Management**
- 4. Career Exploration**
- 5. Career Recommendation**
- 6. Skill-Gap Analysis**
- 7. Personalized Learning Roadmap**
- 8. Resume Builder**
- 9. Job Search**
- 10. Job Application Tracking**
- 11. Interview Preparation**
- 12. Career Chat Assistant**
- 13. Career Progress Dashboard**
- 14. Admin Management**

2. Entities :

- 1. User**
- 2. User Profile**
- 3. Skill**
- 4. User Skill**
- 5. Career**
- 6. Career Skill**
- 7. Career Recommendation**
- 8. Skill Gap**
- 9. Learning Roadmap**
- 10. Roadmap Item**
- 11. Resume**
- 12. Job**
- 13. Job Application**
- 14. Interview Session**
- 15. Interview Question**

3.Attributes :

1. User :-

User ID, Name, Email, Password, Role, Created At, Updated At

2. User Profile :-

Profile ID, User ID, Phone, Date Of Birth, Location, Education, Degree, College, Graduation Year, Experience Years, Career Goal, Interests, Bio, Profile Photo

3. Skill :-

Skill ID, Skill Name, Category, Description

4. User Skill :-

User Skill ID, User ID, Skill ID, Skill Level, Years Of Experience

5. Career :-

Career ID, Career Name, Category, Description, Demand Level, Average Salary, Required Experience

6. Career Skill :-

Career Skill ID, Career ID, Skill ID, Required Level, Importance

7. Career Recommendation :-

Recommendation ID, User ID, Career ID, Match Percentage, Reason, Strengths, Created At

8. Skill Gap :-

Gap ID, User ID, Career ID, Skill ID, Current Level, Required Level, Gap Status

9. Learning Roadmap :-

Roadmap ID, User ID, Career ID, Title, Description, Progress Percentage, Created At, Updated At

10. Roadmap Item :-

Item ID, Roadmap ID, Skill ID, Title, Description, Sequence Number, Estimated Hours, Status

11. Resume :-

Resume ID, User ID, Resume Name, Summary, Education, Experience, Projects, Certifications, Created At, Updated At

12. Job :-

Job ID, Company Name, Job Title, Description, Location, Job Type, Salary, Experience Required, Posted Date, Application URL, Created At

13. Job Application :-

Application ID, User ID, Job ID, Status, Applied Date, Interview Date, Notes

14. Interview Session :-

Session ID, User ID, Career ID, Difficulty, Score, Started At, Completed At

15. Interview Question :-

Question ID, Session ID, Question, Question Type, Difficulty, User Answer, Correct Answer, Score

4. Table Structure :

1. Users

```
mysql> DESC users;
```

Field	Type	Null	Key	Default	Extra
user_id	bigint	NO	PRI	NULL	auto_increment
name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	
password	varchar(255)	NO		NULL	
role	enum('USER','ADMIN')	NO		USER	
created_at	timestamp	NO		CURRENT_TIMESTAMP	DEFAULT_GENERATED
updated_at	timestamp	NO		CURRENT_TIMESTAMP	DEFAULT_GENERATED on update CURRENT_TIMESTAMP

```
7 rows in set (0.07 sec)
```

2. User Profiles

```
mysql> DESC user_profiles;
```

Field	Type	Null	Key	Default	Extra
profile_id	bigint	NO	PRI	NULL	auto_increment
user_id	bigint	NO	UNI	NULL	
phone	varchar(20)	YES		NULL	
date_of_birth	date	YES		NULL	
location	varchar(150)	YES		NULL	
education	varchar(150)	YES		NULL	
degree	varchar(150)	YES		NULL	
college	varchar(200)	YES		NULL	
graduation_year	int	YES		NULL	
experience_years	decimal(4,1)	YES		0.0	
career_goal	varchar(150)	YES		NULL	
interests	text	YES		NULL	
bio	text	YES		NULL	
profile_photo	varchar(500)	YES		NULL	

```
14 rows in set (0.11 sec)
```

3. Skills

```
mysql> DESC skills;
```

Field	Type	Null	Key	Default	Extra
skill_id	bigint	NO	PRI	NULL	auto_increment
skill_name	varchar(100)	NO	UNI	NULL	
category	varchar(100)	NO		NULL	
description	text	YES		NULL	

```
4 rows in set (0.00 sec)
```

4. User Skills

```
mysql> DESC user_skills;
```

Field	Type	Null	Key	Default	Extra
user_skill_id	bigint	NO	PRI	NULL	auto_increment
user_id	bigint	NO	MUL	NULL	
skill_id	bigint	NO	MUL	NULL	
skill_level	enum('BEGINNER','INTERMEDIATE','ADVANCED','EXPERT')	NO		BEGINNER	
years_experience	decimal(4,1)	YES		0.0	

5 rows in set (0.33 sec)

5. Careers

```
mysql> DESC careers;
```

Field	Type	Null	Key	Default	Extra
career_id	bigint	NO	PRI	NULL	auto_increment
career_name	varchar(150)	NO	UNI	NULL	
category	varchar(100)	NO		NULL	
description	text	YES		NULL	
demand_level	enum('LOW','MEDIUM','HIGH','VERY_HIGH')	YES		MEDIUM	
average_salary	decimal(12,2)	YES		NULL	
required_experience	varchar(100)	YES		NULL	

7 rows in set (0.00 sec)

6. Career Skills

```
mysql> DESC career_skills;
```

Field	Type	Null	Key	Default	Extra
career_skill_id	bigint	NO	PRI	NULL	auto_increment
career_id	bigint	NO	MUL	NULL	
skill_id	bigint	NO	MUL	NULL	
required_level	enum('BEGINNER','INTERMEDIATE','ADVANCED','EXPERT')	NO		NULL	
importance	enum('LOW','MEDIUM','HIGH','CRITICAL')	YES		MEDIUM	

5 rows in set (0.00 sec)

7. Career Recommendations

```
mysql> DESC career_recommendations;
```

Field	Type	Null	Key	Default	Extra
recommendation_id	bigint	NO	PRI	NULL	auto_increment
user_id	bigint	NO	MUL	NULL	
career_id	bigint	NO	MUL	NULL	
match_percentage	decimal(5,2)	NO		NULL	
reason	text	YES		NULL	
strengths	text	YES		NULL	
created_at	timestamp	NO		CURRENT_TIMESTAMP	DEFAULT_GENERATED

7 rows in set (0.00 sec)

8. Skill Gaps

```
mysql> DESC skill_gaps;
```

Field	Type	Null	Key	Default	Extra
gap_id	bigint	NO	PRI	NULL	auto_increment
user_id	bigint	NO	MUL	NULL	
career_id	bigint	NO	MUL	NULL	
skill_id	bigint	NO	MUL	NULL	
current_level	enum('NONE','BEGINNER','INTERMEDIATE','ADVANCED','EXPERT')	NO		NONE	
required_level	enum('BEGINNER','INTERMEDIATE','ADVANCED','EXPERT')	NO		NULL	
gap_status	enum('MISSING','NEEDS_IMPROVEMENT','COMPLETED')	NO		MISSING	

7 rows in set (0.00 sec)

9. Learning Roadmaps

```
mysql> DESC learning_roadmaps;
```

Field	Type	Null	Key	Default	Extra
roadmap_id	bigint	NO	PRI	NULL	auto_increment
user_id	bigint	NO	MUL	NULL	
career_id	bigint	NO	MUL	NULL	
title	varchar(200)	NO		NULL	
description	text	YES		NULL	
progress_percentage	decimal(5,2)	YES		0.00	
created_at	timestamp	NO		CURRENT_TIMESTAMP	DEFAULT_GENERATED
updated_at	timestamp	NO		CURRENT_TIMESTAMP	DEFAULT_GENERATED on update CURRENT_TIMESTAMP

8 rows in set (0.00 sec)

10. Roadmap Items;

```
mysql> DESC roadmap_items;
```

Field	Type	Null	Key	Default	Extra
item_id	bigint	NO	PRI	NULL	auto_increment
roadmap_id	bigint	NO	MUL	NULL	
skill_id	bigint	YES	MUL	NULL	
title	varchar(200)	NO		NULL	
description	text	YES		NULL	
sequence_number	int	NO		NULL	
estimated_hours	int	YES		0	
status	enum('NOT_STARTED', 'IN_PROGRESS', 'COMPLETED')	NO		NOT_STARTED	

8 rows in set (0.00 sec)

11. Resumes

```
mysql> DESC resumes;
```

Field	Type	Null	Key	Default	Extra
resume_id	bigint	NO	PRI	NULL	auto_increment
user_id	bigint	NO	MUL	NULL	
resume_name	varchar(150)	NO		NULL	
summary	text	YES		NULL	
education	text	YES		NULL	
experience	text	YES		NULL	
projects	text	YES		NULL	
certifications	text	YES		NULL	
created_at	timestamp	NO		CURRENT_TIMESTAMP	DEFAULT_GENERATED
updated_at	timestamp	NO		CURRENT_TIMESTAMP	DEFAULT_GENERATED on update CURRENT_TIMESTAMP

10 rows in set (0.00 sec)

12. Jobs

```
mysql> DESC jobs;
```

Field	Type	Null	Key	Default	Extra
job_id	bigint	NO	PRI	NULL	auto_increment
company_name	varchar(150)	NO		NULL	
job_title	varchar(150)	NO		NULL	
description	text	YES		NULL	
location	varchar(150)	YES		NULL	
job_type	enum('FULL_TIME', 'PART_TIME', 'INTERNSHIP', 'CONTRACT', 'REMOTE')	YES		FULL_TIME	
salary	decimal(12,2)	YES		NULL	
experience_required	varchar(100)	YES		NULL	
posted_date	date	NO		curdate()	DEFAULT_GENERATED
application_url	varchar(500)	YES		NULL	
created_at	timestamp	NO		CURRENT_TIMESTAMP	DEFAULT_GENERATED

11 rows in set (0.45 sec)

13. Job Applications

```
mysql> DESC job_applications;
```

Field	Type	Null	Key	Default	Extra
application_id	bigint	NO	PRI	NULL	auto_increment
user_id	bigint	NO	MUL	NULL	
job_id	bigint	NO	MUL	NULL	
status	enum('APPLIED','SHORTLISTED','INTERVIEW','SELECTED','REJECTED')	NO		APPLIED	
applied_date	date	NO		curdate()	DEFAULT_GENERATED
interview_date	date	YES		NULL	
notes	text	YES		NULL	

7 rows in set (0.00 sec)

14. Interview Sessions

```
mysql> DESC interview_sessions;
```

Field	Type	Null	Key	Default	Extra
session_id	bigint	NO	PRI	NULL	auto_increment
user_id	bigint	NO	MUL	NULL	
career_id	bigint	NO	MUL	NULL	
difficulty	enum('EASY','MEDIUM','HARD')	NO		MEDIUM	
score	decimal(5,2)	YES		0.00	
started_at	timestamp	NO		CURRENT_TIMESTAMP	DEFAULT_GENERATED
completed_at	timestamp	YES		NULL	

7 rows in set (0.00 sec)

15. Interview Questions

```
mysql> DESC interview_questions;
```

Field	Type	Null	Key	Default	Extra
question_id	bigint	NO	PRI	NULL	auto_increment
session_id	bigint	NO	MUL	NULL	
question	text	NO		NULL	
question_type	enum('TECHNICAL','HR','BEHAVIORAL','CODING')	NO		NULL	
difficulty	enum('EASY','MEDIUM','HARD')	NO		NULL	
user_answer	text	YES		NULL	
correct_answer	text	YES		NULL	
score	decimal(5,2)	YES		0.00	

8 rows in set (0.00 sec)

5. ER Diagram :-

